

Note on Quantum Hall Effect

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Abstract

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1 Preparations

First we'll give some preparations for the discussion of the QHE. We'll start with some basic concepts in CMT and QM. I won't give a detailed explanation, yet just list out some important results as a reference for later use.

1.1 Band Theory and Conductivity

1.2 Landau Levels

1.3 Berry Phase

2 Integer Quantum Hall Effect

2.1 Experimental Phenomenon

Bibliography